TTex Employee Fraud Analysis

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1. Executive Summary

This analysis focuses on analyzing company vehicle usage, trip distances, and fuel consumption to identify potential irregularities and assess data quality. Data from 64 files was cleaned, validated, and merged into a single dataset. The analysis included verifying employee and vehicle activity periods, examining trip distances, and evaluating fuel consumption patterns.

The analysis identified 30 trips with potentially unusual distances and significant fuel consumption discrepancies, particularly involving two employees from the Transport Department using a Fiat Ducato and a Ford Transit.

The findings highlight potential data quality issues and possible overreporting of fuel consumption that require further investigation.

2. Problem And Objectives

The company needs to ensure the accuracy of its vehicle usage records and identify potential irregularities in reported trip distances and fuel consumption.

The main objectives of the analysis were to: - Trip distance analysis: Identify unusual distances between the same locations and detect potentially inaccurate trip records. - Fuel consumption analysis: Evaluate fuel efficiency across vehicles and identify unusually high reported fuel consumption. - Employee activity verification: Ensure that trips were made by active employees using active company vehicles. - Risk identification: Highlight potentially suspicious trips, vehicles, and employees requiring further investigation.

The ultimate goal is to identify potential operational inefficiencies, and support better control of company vehicle usage and fuel expenses.

3. Data Validatiion And Peparation

Data from all weeks (64 files in total) was combined into five separate DataFrames: df_department, df_position, df_employee, df_car and df_trip. Duplicates resulting from the concatenation of the files were removed. The correctness of the data types was verified. After data cleaning, the suffix "_clean" was added to the names of all DataFrames.

1. The **df_department** DataFrame contains 5 unique records, each representing an organizational department and consisting of three fields: id, name, and description.
2. The **df_position** DataFrame contains 36 unique records, the IDs and names of job positions within the company.
3. The **df_employee** DataFrame contains 36 unique records with employee information, including first names, last names, gender, department IDs, position IDs, dates of birth, hire dates, and employment termination dates.
4. The **df_car** DataFrame contains 20 unique records related to vehicles used within the company. It includes the following columns: car ID, brand, model, year of manufacture, fuel type, transport usage, activity status, activity start date, and activity end date. Missing values were identified in the activity status column and filled with appropriate values based on the activity dates.
5. The **df_trip** DataFrame contains trip IDs, start and end dates, car IDs, employee IDs, distance traveled, fuel consumption, and start and end locations. The DataFrame contained no duplicate records. Incorrect values in the from and to columns were corrected. After data cleaning, 13 unique locations remained in these columns, including cities and other places, such as offices and shops.

Finally, all cleaned DataFrames were merged into a single DataFrame named df_total, as shown below.

Table 1: Total DataFrame Structure

column	dtype
trip_id	int64
trip_start	datetime64[us]
trip_end	datetime64[us]
trip_distance	int64
trip_fuel	float64
trip_from	str
trip_to	str
car_brand	str
car_model	str
car_year	int64
car_fuel_type	str
car_transport	bool
car_active	str
car_active_from	datetime64[us]
car_active_to	datetime64[us]
employee_name	str
employee_surname	str
employee_gender	str
employee_year_of_birth	int64

column	dtype
employee_date_of_employment	datetime64[us]
employee_employment_end_date	datetime64[us]
employee_position_name	str
employee_department_name	str
employee_department_description	str

```
Shape: (4432, 24)
```

4. Exploratory Data Analysis

A total of 443 trips were completed. Trip **distances** ranged from 3 km (between Olsztyn and the shop or office) to 576 km (between Szczecin and Olsztyn), with an average distance of 150.34 km.

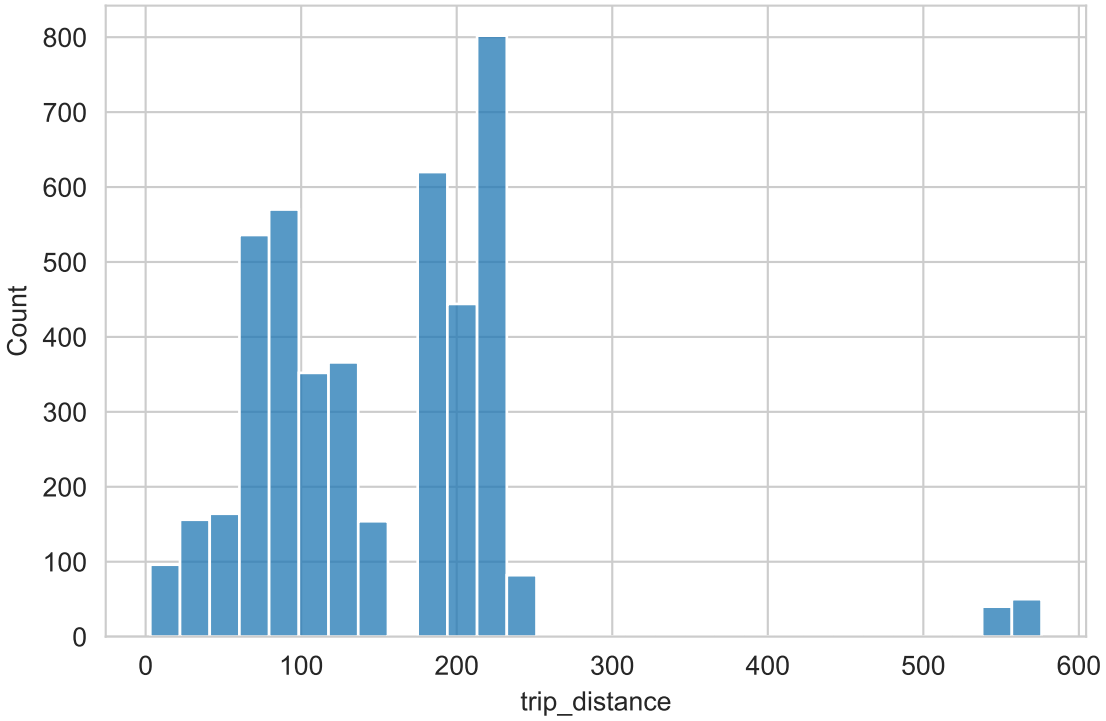

Figure 1: Distribution of Trip Distances

The **trip dates were validated and found to be consistent**. In all cases, the trip start date was earlier than or equal to the end date.

No **trips involving inactive vehicles were detected**, i.e., trips that started after the vehicle's activity end date or started before that date but ended after it.

No trips made by employees before their employment start date or after their employment end date were detected.

Most intercity routes show a very similar relative range of distances, approximately 7–8%.

Table 2: Descriptive Statistics of Trip Distances by Route

route	count	mean	std	min	25%	50%	75%	max
Bartoszyce Olsztyn	536.00	73.53	1.61	71.00	72.00	73.50	75.00	76.00
Bydgoszcz Olsztyn	544.00	219.54	4.60	212.00	215.00	220.00	223.00	228.00
Elbląg Olsztyn	378.00	113.92	2.44	110.00	112.00	114.00	116.00	118.00
Gdańsk Olsztyn	486.00	197.39	4.02	190.00	194.00	197.00	201.00	204.00
Malbork Olsztyn	494.00	134.87	2.77	130.00	133.00	135.00	137.00	140.00
Mikołajki Olsztyn	570.00	87.09	1.82	84.00	85.00	87.00	89.00	90.00
Olsztyn Ostróda	320.00	41.52	0.90	40.00	41.00	42.00	42.00	43.00
Olsztyn Sklep	38.00	4.58	1.62	3.00	3.00	4.00	5.00	8.00
Olsztyn Szczecin	90.00	557.58	11.86	538.00	547.00	560.00	566.00	576.00
Olsztyn Toruń	510.00	186.36	3.78	180.00	183.00	186.00	189.00	193.00
Olsztyn Urząd	58.00	6.41	2.36	4.00	4.00	7.00	7.00	11.00
Olsztyn Warszawa	408.00	228.27	4.70	220.00	224.00	228.50	232.00	236.00

The median distance was calculated for each route, and trips exceeding the median distance by more than 10% were identified. A total of 30 potentially suspicious trips were identified.

Table 3: Suspicious Trips

route	distance	Mdn	Mdn_dev	E_name	E_surname	count
Olsztyn Sklep	8	4.00	100.00	Mariola	Suchań	2
Olsztyn Sklep	7	4.00	75.00	Fryderyk	Przybylski	2
Olsztyn Sklep	7	4.00	75.00	Jadwiga	Gall	4
Olsztyn Urząd	11	7.00	57.14	Ernest	Rząca	2
Olsztyn Urząd	11	7.00	57.14	Roksana	Czachor	2
Olsztyn Urząd	10	7.00	42.86	Roksana	Czachor	2
Olsztyn Urząd	10	7.00	42.86	Fryderyk	Przybylski	2
Olsztyn Urząd	10	7.00	42.86	Juliusz	Ikonowicz	4
Olsztyn Sklep	5	4.00	25.00	Fryderyk	Przybylski	2
Olsztyn Sklep	5	4.00	25.00	Emil	Sekula	6
Olsztyn Sklep	5	4.00	25.00	Ernest	Rząca	2

Legend: Mdn - median, Mdn_dev - distance deviation pct, E - employee.

The **average fuel consumption per 100 km** was calculated for each vehicle used. The minimum and maximum values differ significantly from the mean and median, which may indicate substantial inaccuracies in the reported fuel consumption data. **A threshold of 50% above the median fuel consumption for each vehicle** was adopted as an indicator of potentially incorrect fuel consumption values for individual trips. As a result, **670 trips with potential irregularities were identified**.

Table 4: Fuel Consumption per 100 km per car

car_brand	car_model	car_year	car_fuel_type	count	mean	median	std	min	max
Fiat	Ducato	2011	ON	1210	11.05	9.53	4.55	7.87	27.44
		2017	ON	614	11.02	9.14	4.52	7.87	27.44
Ford	Transit	2013	ON	1126	12.57	10.22	5.68	8.80	30.71
		2017	ON	1196	12.49	10.65	5.34	8.67	30.71
Opel	Insignia	2021	PB	286	7.57	7.65	0.39	5.42	7.75

Two employees were identified as significantly overreporting fuel consumption. These employees traveled in a Fiat Ducato and a Ford Transit on nearly all routes. In the table, Avg_dev_% represents the average percentage deviation in fuel consumption from the median, calculated across all trips made by each employee, including those without detected irregularities.

Table 5: Suspicious Employees

E_name	E_surname	department	suspicious_trips	Avg_dev_%
Magdalena	Dobrzyńska	Transport	544	102.77
Emil	Sekula	Transport	126	30.78

5. Recomendation

First, discrepancies in trip distances that deviated significantly from the median should be investigated. Most of these involved trips to offices and shops, where the destinations were not precisely defined. Therefore, these differences do not necessarily indicate irregularities.

The significantly higher fuel consumption reported by employees in the Transport Department also requires further investigation. Fuel consumption levels nearly twice the expected values strongly suggest that the reported amount of fuel used may have been substantially overestimated.